Provenance of 'Deflating degrees of freedom: dirac16complex when x_5, x_6, x_7 deflate' and of deflation.nb

1. The request

The user's instructions of 2026-10-30, verbatim:

```
next, study the case where x5, x6, and x7 are associated to metric components (call them g55, g66, g77) that each
     exponentially deflate (eg., g55 = b55*Exp[ - a*x4 ], where a>0 and b55>0. call   x5, x6, and x7  deflating
     degrees of freedom. Document your results, in entirety, in a new mathematica notebook named "deflation.nb".

'where x5, x6, and x7 are associated to metric components' means g55* (d x5 )^2 +g66* (d x6 )^2 +g77* (d x7 )^2 .
```

In the project x_5, x_6, x_7 are timelike. Asked whether they should stay timelike, the user answered:

```
-g55* (d x5 )^2 -g66* (d x6 )^2 -g77* (d x7 )^2 ; good catch of my error
```

and, asked what the remaining metric components should be:

```
Primordial field of 6planes.nb
```

Hence the line element used everywhere, with $z = 6Hx_0$:

$$ds^2 = \tan^2 z\, dx_0^2 + e^{-2Hx_4} \sin^{-1/3} z\, \left(dx_1^2 + dx_2^2 + dx_3^2\right) - dx_4^2 - g_{55}\, dx_5^2 - g_{66}\, dx_6^2 - g_{77}\, dx_7^2,$$
$$g_{jj} = b_{jj}\, e^{-ax_4}\ (j = 5, 6, 7), \quad a > 0, \quad b_{jj} > 0.$$

The standing rules of the project were followed: every document has a provenance file and a symbols file, each in Markdown, LaTeX and PDF, and each file contains every equation it uses.

2. Sources

- **Pre-gravity6plane.nb** (P. L. Nash, (C) 2022, GNU GPL). Used: the coordinates $X = \{x_0, \ldots, x_7\}$ (cell 45); $\eta = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$ (cell 66); $\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$ (cell 73); τ_A and $\bar{\tau}_A = \sigma(\sigma\tau_A)^T$ (cell 265), built from the self-dual and anti-self-dual 4×4 matrices and the six antisymmetric 8×8 matrices of the notebook; $T^A = (T16^A)[A] = \begin{pmatrix} 0 & \bar{\tau}_A \\ \tau_A & 0 \end{pmatrix}$ (cell 308); $\sigma_{16} = T^0T^1T^2T^3$ (cells 306, 333); the name "wave function for a Universe" for a mode function (cell 78); the vielbein of text cells 901 and 913, $\mathrm{diag}\left(\tan 6Hx_0,\ e^{-a_4(Hx_4)} \sin^{-1/6} 6Hx_0\ (\times 3),\ 1,\ e^{a_4(Hx_4)} \sin^{-1/6} 6Hx_0\ (\times 3)\right)$.
- **The primordial gravitational field of 6planes.nb** (this repository, 2026-10-30), chosen by the user: the vielbein above with $a_4(Hx_4) = Hx_4$, $\mathrm{diag}\left(\tan z,\ e^{-Hx_4} \sin^{-1/6} z\ (\times 3),\ 1,\ e^{Hx_4} \sin^{-1/6} z\ (\times 3)\right)$. In deflation.nb its last three entries are replaced by $e_j = \sqrt{b_{jj}}\, e^{-ax_4/2}$, so that $e_j^2 = g_{jj}$.
- **Code reused verbatim** from `code/6planes_src.txt`, the source of 6planes.nb: the construction of T^A (Part 1.1), the exact Grassmann-algebra engine (Part 2), and the curved-space tools `spinConnection`, `covConst`, `christoffelG`, `riemannG`, `ricciG`, `einsteinG`, `kretschmannG` (Part 3). All other cells of deflation.nb were written for this work.
- **Standard results used** (textbook statements, not computed here):
  - Grassmann (Berezin) calculus with left derivatives and the order-reversing conjugation $(\theta\zeta)^* = \zeta^*\theta^*$ [F. A. Berezin, The Method of Second Quantization, Academic Press (1966)].
  - The tetrad (vielbein) formalism for spinors in curved space, and particle creation by time-dependent gravitational fields [L. Parker, Phys. Rev. 183, 1057 (1969); N. D. Birrell and P. C. W. Davies, Quantum Fields in Curved Space, Cambridge University Press (1982)].
  - The Poincare upper-half-space model $\frac{r^2}{w^2}(dw^2 + dy_5^2 + dy_6^2 + dy_7^2)$ of the hyperbolic 4-space of radius r, of constant sectional curvature $-1/r^2$ [J. G. Ratcliffe, Foundations of Hyperbolic Manifolds, Springer (1994)].
  - Indefinite (Krein) inner products [J. Bognar, Indefinite Inner Product Spaces, Springer (1974)].
  - The mean-field (Kohn-Sham) effective mass $m + \lambda\rho_s$ [W. Kohn and L. J. Sham, Phys. Rev. 140, A1133 (1965)].

3. Computations

All results are outputs of deflation.nb, produced with Wolfram Language 15.0.1 for Mac OS X ARM (64-bit) on 2026-10-30. The notebook is generated from `code/deflation_src.txt` (the concatenation of `code/deflation_head.txt` and `code/deflation_tail.txt`) by `code/build_deflation.wls`. The builder evaluates every input cell in the kernel with `EvaluationData` and writes the inputs, every message and every output into deflation.nb: 17 input cells, 0 messages, 28 s (log: `code/build_deflation.log`). Exact simplifications use the assumptions

$$H>0, \qquad 0<x_0<\frac{\pi}{12H}, \qquad a>0, \qquad b_{55},b_{66},b_{77}>0, \qquad x_0,\dots,x_7\in\mathbb{R}, \qquad m,\lambda\in\mathbb{R}.$$

3.1 Results and where they are computed

Part 0, Out[1]. The diagonal of η is $(1,1,1,1,-1,-1,-1,-1)$.

Part 1.1, Out[2]. The eight matrices T^A, displayed.

Part 1.2, Out[3] (all True). The entries of T^A are in $\{-1,0,1\}$; $T^0,\dots,T^3$ are symmetric and $T^4,\dots,T^7$ antisymmetric; and

$$T^AT^B+T^BT^A=2\eta^{AB}I_{16}, \qquad \Gamma=T^0\cdots T^7=\mathrm{diag}(-I_8,I_8), \qquad \sigma_{16}=\mathrm{diag}(-\sigma,\sigma),$$
$$(\sigma_{16}T^A)^T=-\sigma_{16}T^A, \qquad [S^{AB},T^C]=\eta^{BC}T^A-\eta^{AC}T^B, \qquad S^{AB}=\tfrac{1}{4}[T^A,T^B].$$

Part 2, Out[4] (all True). $\psi_1\psi_2+\psi_2\psi_1=0$, $\psi_1^2=0$, $(\psi_1^*\psi_1)^*=\psi_1^*\psi_1$.

Part 3, Out[6] (all True). For $K=-i\sigma_{16}T^4$, $\beta=-iT^4$ and $\Pi_\pm=\frac{1}{2}(I_{16}\pm K)$: $K^\dagger=K$; $K^2=I_{16}$; the eigenvalue $+1$ occurs 8 times and -1 occurs 8 times; $KT^A=T^AK$ $(A=0..4)$; $KT^A=-T^AK$ $(A=5,6,7)$; $\sigma_{16}=K\beta$. For $A=0..4$, and also without T^A:

$$\Pi_+\sigma_{16}T^A\Pi_-=0, \qquad \Pi_+\sigma_{16}T^A\Pi_+=\Pi_+\beta T^A\Pi_+, \qquad \Pi_-\sigma_{16}T^A\Pi_-=-\Pi_-\beta T^A\Pi_-.$$

For $\psi=e^{-i\omega x_4}u$ with arbitrary complex u:

$$\tfrac{1}{2}\left(\bar\psi T^4\partial_4\psi-(\partial_4\bar\psi)T^4\psi\right)=\omega\,\psi^\dagger K\psi.$$

Part 4, Out[7]. With $e_0=\tan z$, $e_{1,2,3}=e^{-Hx_4}\sin^{-1/6}z$, $e_4=1$, $e_j=\sqrt{b_{jj}}\,e^{-ax_4/2}$ and $g_{\mu\nu}=\eta_{AB}e^A{}_\mu e^B{}_\nu$: $e_j^2=g_{jj}$ (True); $(\sqrt{|g|})^2=|\det g_{\mu\nu}|$ (True); the entries of $g_{\mu\nu}$ for $\mu=5,6,7$ equal $-g_{55},-g_{66},-g_{77}$ (True);

$$g_{\mu\nu}=\mathrm{diag}\left(\tan^2z,\ e^{-2Hx_4}\sin^{-1/3}z\ (\times3),\ -1,\ -b_{55}e^{-ax_4},\ -b_{66}e^{-ax_4},\ -b_{77}e^{-ax_4}\right),$$

$$\sqrt{g_{55}}=\sqrt{b_{55}e^{-ax_4}}, \qquad \frac{d\ln\sqrt{g_{55}}}{dx_4}=-\frac{a}{2}, \qquad \sqrt{g_{55}g_{66}g_{77}}=\sqrt{b_{55}b_{66}b_{77}\,e^{-3ax_4}},$$
$$\sqrt{|g|}=\sec z\,\sqrt{b_{55}b_{66}b_{77}\sin z}\;e^{-3(a+2H)x_4/2}.$$

Part 5, Out[8]. The full deflation field, with the Christoffel symbols, Riemann, Ricci and Einstein tensors

$$\Gamma^\lambda{}_{\mu\nu}=\tfrac{1}{2}g^{\lambda\rho}(\partial_\mu g_{\rho\nu}+\partial_\nu g_{\rho\mu}-\partial_\rho g_{\mu\nu}), \qquad R^\kappa{}_{\lambda\mu\nu}=\partial_\mu\Gamma^\kappa{}_{\lambda\nu}-\partial_\nu\Gamma^\kappa{}_{\lambda\mu}+\Gamma^\kappa{}_{\mu\rho}\Gamma^\rho{}_{\lambda\nu}-\Gamma^\kappa{}_{\nu\rho}\Gamma^\rho{}_{\lambda\mu},$$

$$R_{\lambda\nu}=R^\kappa{}_{\lambda\kappa\nu}, \qquad R=g^{\lambda\nu}R_{\lambda\nu}, \qquad G_{\mu\nu}=R_{\mu\nu}-\tfrac{1}{2}Rg_{\mu\nu}, \qquad R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu}=\sum\frac{g_{\kappa\kappa}}{g_{\lambda\lambda}g_{\mu\mu}g_{\nu\nu}}(R^\kappa{}_{\lambda\mu\nu})^2,$$

gives

$$R=\frac{3}{8}\left[3(a^2+3aH-32H^2)-4(a^2+3aH+32H^2)\cos2z+a(a+3H)\cos4z\right]\csc^4z,$$

$$G^0{}_0=-\frac{3}{16}\left[(a^2+3aH+2H^2)(3+\cos4z)-4(a^2+3aH+6H^2)\cos2z\right]\csc^4z,$$

$$G^i{}_i=-\frac{3}{16}\left[3a^2+6aH-64H^2-4(a^2+2aH+20H^2)\cos2z+a(a+2H)\cos4z\right]\csc^4z \quad (i=1,2,3),$$

$$G^4{}_4 = \frac{3}{4}(a^2+4H^2) - \frac{3}{16}\left[3(a^2+3aH-32H^2) - 4(a^2+3aH+32H^2)\cos 2z + a(a+3H)\cos 4z\right]\csc^4 z$$
$$= \frac{3}{4}(a^2+4H^2) - \frac{R}{2},$$

$$G^j{}_j = -\frac{3}{32}\left[3(a^2+4aH-64H^2) - 4(a^2+4aH+64H^2)\cos 2z + a(a+4H)\cos 4z\right]\csc^4 z \quad (j=5,6,7),$$

$$R_{04} = -3H^2\cot z \text{ (the only non-zero } R_{\mu\nu},\ \mu<\nu), \qquad G_{04} = R_{04} \text{ (True)},$$

$$R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = \frac{3}{2}\left[a^4 + 6a^2H^2 + 8H^4 + H^4(1245 + 108\cos 2z - \cos 4z)\cot^4 z\csc^4 z + 8H^4\cos^2 2z\ \csc^8 z\right].$$

R, $G^\mu{}_\mu$, R_{04} and the Kretschmann scalar do not contain x_4 (True), and

$$\lim_{x_0\to 0^+} x_0^4 R = -\frac{7}{108H^2}, \qquad \lim_{x_0\to 0^+} x_0^8\, R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = \frac{85}{69984H^4}.$$

Part 5, Out[9]. The pure deflation field $ds^2 = dx_0^2 + dx_1^2 + dx_2^2 + dx_3^2 - dx_4^2 - g_{55}dx_5^2 - g_{66}dx_6^2 - g_{77}dx_7^2$ gives $R = 3a^2$, $R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = 3a^4/2$, $G^\mu{}_\mu = -3a^2/2$ ($\mu = 0..3$) and $G^\mu{}_\mu = -3a^2/4$ ($\mu = 4..7$).

Part 5, Out[10] (all True). Every Riemann component of the pure deflation field with an index in $\{0,1,2,3\}$ vanishes. On the block (x_4,x_5,x_6,x_7), $R_{\kappa\lambda\mu\nu} = \frac{a^2}{4}(g_{\kappa\mu}g_{\lambda\nu} - g_{\kappa\nu}g_{\lambda\mu})$ with $R_{\kappa\lambda\mu\nu} = g_{\kappa\rho}R^\rho{}_{\lambda\mu\nu}$. With $x_4 = \frac{2}{a}\ln\frac{aw}{2}$ and $x_j = y_j/\sqrt{b_{jj}}$, the metric of the block is $-\frac{4}{a^2w^2}I_4$ in the coordinates (w,y_5,y_6,y_7), that is

$$-dx_4^2 - g_{55}dx_5^2 - g_{66}dx_6^2 - g_{77}dx_7^2 = -\frac{4}{a^2}\,\frac{dw^2+dy_5^2+dy_6^2+dy_7^2}{w^2}.$$

Part 6, Out[11]. For $p_5 = -g_{55}\dot{x}_5$, $\dot{} = d/d\ell$, and the normalization $-\dot{x}_4^2 - g_{55}\dot{x}_5^2 = -1$:

$$\dot{x}_5 = -\frac{e^{ax_4}p_5}{b_{55}}, \qquad \dot{x}_4^2 = 1 - \frac{e^{ax_4}p_5^2}{b_{55}}, \qquad x_4^* = \frac{1}{a}\ln\frac{b_{55}}{p_5^2}.$$

The following are True: $\frac{1}{2}\,d(\dot{x}_4^2)/dx_4 = -\Gamma^4{}_{55}\dot{x}_5^2$; with $w = \frac{2}{a}e^{ax_4/2}$, $y_5 = \sqrt{b_{55}}x_5$ and $w_* = 2\sqrt{b_{55}}/(ap_5)$ ($p_5>0$), $(dw/dy_5)^2 = (w_*^2 - w^2)/w^2$; and $w(x_4^*) = w_*$.

Part 7, Out[12]. With $\gamma^\mu = E_A{}^\mu T^A$, $D_\mu\psi = \partial_\mu\psi + \Omega_\mu\psi$, $\Omega_\mu = \frac{1}{4}\omega_\mu{}^A{}_B T_A T^B$, $T_A = \eta_{AB}T^B$, $\omega_\mu{}^A{}_B = e^A{}_\nu(\partial_\mu E_B{}^\nu + \Gamma^\nu{}_{\mu\lambda}E_B{}^\lambda)$ and

$$\mathcal{L} = \sqrt{|g|}\left[\tfrac{1}{2}\left(\bar\psi\gamma^\mu D_\mu\psi - (D_\mu\bar\psi)\gamma^\mu\psi\right) - m\bar\psi\psi - \frac{\lambda}{2}(\bar\psi\psi)^2\right], \qquad D_\mu\bar\psi = \partial_\mu\bar\psi - \bar\psi\Omega_\mu,$$

the following are True:

- $\partial_\mu\gamma^\nu + \Gamma^\nu{}_{\mu\lambda}\gamma^\lambda + [\Omega_\mu,\gamma^\nu] = 0$;
- $\mathcal{L}^* = \mathcal{L}$;
- $\partial^L\mathcal{L}/\partial\psi_a^* - \partial_\mu\,\partial^L\mathcal{L}/\partial(\partial_\mu\psi_a^*) = \left[\sqrt{|g|}\sigma_{16}(\gamma^\mu D_\mu\psi - m\psi - \lambda(\bar\psi\psi)\psi)\right]_a$ for all 16 components;
- $\mathcal{L}_m[\Gamma\psi^*] = \mathcal{L}_{-m}[\psi]$ and $\mathcal{L}_m[T^1\psi(R_1x)] = \mathcal{L}_{-m}[\psi](R_1x)$.

The computed values are

$$E_A{}^A = \left(\cot z,\ e^{Hx_4}\sin^{1/6} z\ (\times 3),\ 1,\ e^{ax_4/2}/\sqrt{b_{55}},\ e^{ax_4/2}/\sqrt{b_{66}},\ e^{ax_4/2}/\sqrt{b_{77}}\right),$$
$$\gamma^\mu\Omega_\mu = -\frac{3}{2}H\cot^2 z\,T^0 - \frac{3}{4}(a+2H)T^4.$$

Part 8, Out[13]. True: $\gamma^\mu\partial_\mu F + F\gamma^\mu\Omega_\mu = 0$ for $F = \sin^{1/4} z\, e^{3(a+2H)x_4/4}$; $S^2 = \sum_{A\neq 4}\eta_{AA}(c_Ak_A)^2 I_{16}$ and $ST^4 + T^4S = 0$ for $S = \sum_{A\neq 4} c_Ak_AT^A$, $c_A = 1/e_A$, $(k_A) = (k_0,k_1,k_2,k_3,0,q_5,q_6,q_7)$. The dispersion relation is

$$\omega^2 = m^2 - e^{ax_4}\left(\frac{q_5^2}{b_{55}} + \frac{q_6^2}{b_{66}} + \frac{q_7^2}{b_{77}}\right) + k_0^2\cot^2 z + e^{2Hx_4}(k_1^2+k_2^2+k_3^2)\sin^{1/3} z.$$

With $\bar q^{\,2} = q_5^2/b_{55} + q_6^2/b_{66} + q_7^2/b_{77}$ the following are True: for $k_1 = k_2 = k_3 = 0$, $\omega^2 = m^2 + k_0^2 \cot^2 z - e^{a x_4} \bar q^{\,2}$ and $d\omega^2/dx_4 = -a e^{a x_4} \bar q^{\,2}$; $\omega^2 = 0$ at $x_4^* = \frac{1}{a} \ln[(m^2 + k_0^2 \cot^2 z)/\bar q^{\,2}]$; and at $a = 2H$, $\omega^2 = m^2 + k_0^2 \cot^2 z + C_{2H} e^{2H x_4}$ with the computed coefficient $C_{2H} = (k_1^2 + k_2^2 + k_3^2) \sin^{1/3} z - \bar q^{\,2}$. The exact limits $\lim_{x_4 \to \infty} \omega^2$ for $H = 1$, $x_0 = 1/10$, $m = 1$, $k_0 = 0$, $k_1 = 1$, $k_2 = k_3 = 0$, $q_5 = 1$, $q_6 = q_7 = 0$, $b_{55} = b_{66} = b_{77} = 1$ are $-\infty$ ($a = 3$), $-\infty$ ($a = 2$) and $+\infty$ ($a = 1$).

Part 8, Out[14]. True: in the pure deflation field γ^μ is covariantly constant and $\gamma^\mu \Omega_\mu = -\frac{3a}{4} T^4$; and for $\psi = e^{3a x_4/4} e^{i(k_1 x_1 + q_5 x_5)} \chi(x_4)$ with arbitrary χ,

$$\gamma^\mu D_\mu \psi - m\psi = e^{3a x_4/4} e^{i(k_1 x_1 + q_5 x_5)}\, T^4 (\chi' - M\chi), \qquad M = T^4 \left(i k_1 T^1 + i q_5 \frac{e^{a x_4/2}}{\sqrt{b_{55}}} T^5 - m \right).$$

The numbers are listed in Section 3.2.

Part 8, Out[15]. A figure with three panels: $N = \chi^\dagger \chi$ on a logarithmic scale with a dashed line at $x_4^* = \ln 8$; P_-/N; and P_+, P_- on a logarithmic scale.

Part 9, Out[16]. The matrix $(I_{16} + K)\sigma_{16} c_A T^A (I_{16} - K)$ vanishes for $A = 0, \dots, 4$ (True) and not for $A = 5, 6, 7$ (False, as expected). For $A = 5, 6, 7$ the absolute values of its entries are 0 and $2e^{a x_4/2}/\sqrt{b_{jj}}$.

Part 9, Out[17] (all True). For real symbols κ_A ($A \neq 4$) and $M = T^4 (i \sum_{A \neq 4} \kappa_A T^A - m)$:

$$\sigma_{16} T^4 = iK, \qquad M^\dagger K + K M = 0, \qquad M^\dagger + M = 2 i T^4 (\kappa_5 T^5 + \kappa_6 T^6 + \kappa_7 T^7), \qquad K(M^\dagger + M) + (M^\dagger + M) K = 0.$$

3.2 The numerical integration

The system $\chi' = T^4 (i k_1 T^1 + i q_5 e^{a x_4/2} T^5/\sqrt{b_{55}} - m)\chi$ with $a = b_{55} = k_1 = m = 1$, $q_5 = 1/2$ was solved by the Wolfram Language function `NDSolve` on $-8 \le x_4 \le 5$ with `PrecisionGoal -> 10`, `AccuracyGoal -> 10`, `MaxSteps -> 10^6`. The initial vector $\chi(-8)$ is the normalized eigenvector of $\Pi_+ h_0 \Pi_+ + 10\,\Pi_-$, with $h_0 = -k_1 T^4 T^1 - i m T^4$, whose eigenvalue is closest to $\sqrt{k_1^2 + m^2} = \sqrt{2}$. Because h_0 commutes with K, this matrix acts as h_0 on the $K = +1$ sector and as 10 on the $K = -1$ sector, so $\chi(-8)$ is an eigenvector of h_0 with eigenvalue $\sqrt{2}$ in the $K = +1$ sector. The notebook confirms $\chi^\dagger h_0 \chi = 1.4142135623730951$ and $\chi^\dagger \Pi_- \chi = 0$. Outputs, with $N = \chi^\dagger \chi$ and $P_\pm = \chi^\dagger \Pi_\pm \chi$:

- $x_4^* = \ln 8 = 2.0794415416798357$.
- N at $x_4 = -8, 0, x_4^*, 3, 4, 5$: 1, 1.067409453695482, 1.9672504379730684, 10.010160722635687, 1198.1900459703008, $1.0587953046760937 \times 10^7$.
- P_-/N at $x_4 = -8, 0, x_4^*, 3, 5$: $1.232595164407831 \times 10^{-31}$, 0.03157622876752146, 0.24583816482812526, 0.45005076290732215, 0.499999952776518.
- P_+ at the same points: 1, 1.033704688596978, 1.4836252005444437, 5.505080252588585, $5.293977023380475 \times 10^6$.
- P_- at the same points: $1.232595164407831 \times 10^{-31}$, 0.03370476509850364, 0.4836252374286248, 4.505080470047102, $5.293976023380458 \times 10^6$.
- $Q = P_+ - P_-$ at the same points: 1, 0.9999999234984744, 0.999999963115819, 0.999999782541483, 1.0000000167638063. The largest deviation from the exact value 1 is 2.18×10^{-7}, the integration error.
- $\max_{x_4 = -8, -7, \dots, 5} |P_-/N - (N-1)/(2N)| = 7.17543257486861 \times 10^{-7}$.

The guide quotes these numbers rounded, for example 1.067409, 0.245838, 5.293977×10^6 and 0.49999995; the value $\sin^{1/3}(3/5) = 0.8265$ quoted with the limits was evaluated directly.

4. Derivations in the guide that combine computed identities

- $\Pi_- \sigma_{16} T^A \Pi_+ = 0$ is the Hermitian conjugate of $\Pi_+ \sigma_{16} T^A \Pi_- = 0$, using $(\sigma_{16} T^A)^T = -\sigma_{16} T^A$, $\sigma_{16}^T = \sigma_{16}$ and $\Pi_\pm^\dagger = \Pi_\pm$.
- $(X + m)(X - m) = (\omega^2 - s^2 - m^2) I_{16}$ for $X = iS - i\omega T^4$ follows from $ST^4 + T^4 S = 0$, $S^2 = s^2 I_{16}$ and $(T^4)^2 = -I_{16}$; so non-zero u with $(X - m)u = 0$ exist exactly when $\omega^2 = m^2 + s^2$.
- A frozen-coefficient mode obeys $\chi' = M\chi$ with $\kappa_A = c_A k_A$: from $iS\chi + T^4 \partial_4 \chi = m\chi$ and $(T^4)^{-1} = -T^4$, $\partial_4 \chi = T^4 (iS - m)\chi$.

- $\frac{d}{dx_4}(\chi^\dagger K\chi) = \chi^\dagger(M^\dagger K + KM)\chi = 0$ and $\frac{d}{dx_4}(\chi^\dagger\chi) = \chi^\dagger(M^\dagger + M)\chi$; with Q constant, $P_\pm = (N \pm Q)/2$ and $dP_+/dx_4 = dP_-/dx_4 = \frac{1}{2}dN/dx_4$.
- $\bar{\psi}\gamma^4\psi = \psi^\dagger\sigma_{16}T^4\psi = i\psi^\dagger K\psi$ because ${E_4}^4 = 1$ and $\sigma_{16}T^4 = iK$. In the pure deflation field $\sqrt{|g|} = \sqrt{b_{55}b_{66}b_{77}}\, e^{-3ax_4/2}$, so $\sqrt{|g|}\psi^\dagger K\psi = \sqrt{b_{55}b_{66}b_{77}}\,\chi^\dagger K\chi$ for $\psi = e^{3ax_4/4}e^{i(k_1x_1+q_5x_5)}\chi$.
- $R = 12/(2/a)^2$ and $R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = 24/(2/a)^4$ are the values for a 4-dimensional space of constant curvature with $R_{\kappa\lambda\mu\nu} = \frac{a^2}{4}(g_{\kappa\mu}g_{\lambda\nu} - g_{\kappa\nu}g_{\lambda\mu})$: $R = 4\cdot 3\cdot\frac{a^2}{4}$ and $R_{\kappa\lambda\mu\nu}R^{\kappa\lambda\mu\nu} = 2\cdot 4\cdot 3\cdot\left(\frac{a^2}{4}\right)^2$.
- The semicircles $(y_5 - y_{5c})^2 + w^2 = w_*^2$ solve $(dw/dy_5)^2 = (w_*^2 - w^2)/w^2$, since $dw/dy_5 = -(y_5 - y_{5c})/w$ on them.
- The friction-free form $\gamma^\mu D_\mu(F\chi) = F\gamma^\mu\partial_\mu\chi$ follows from $\gamma^\mu D_\mu(F\chi) = (\gamma^\mu\partial_\mu F + F\gamma^\mu\Omega_\mu)\chi + F\gamma^\mu\partial_\mu\chi$.

5. New in this work

The deflation field and its proper sizes; its exact Ricci, Einstein and Kretschmann curvature and their independence of x_4; the identification of the pure deflation block (x_4, x_5, x_6, x_7) with an all-timelike hyperbolic 4-space of radius $2/a$; the turning point and semicircle geodesics; the field equation of dirac16complex in the deflation field; the exact friction-removing rescaling F; the dispersion relation with its turning point and the critical deflation rate $a_c = 2H$; the exact first-order equation $\chi' = M\chi$ and its numerical solution; and the pair-creation theorem from the conservation of the Krein charge.

6. Typesetting

The Markdown files are converted to LaTeX by `code/md2tex.py` (Python 3.14.7) and compiled with pdfTeX 3.141592653-2.6-1.40.29 (TeX Live 2026).